Quantum mechanics of gravitational waves

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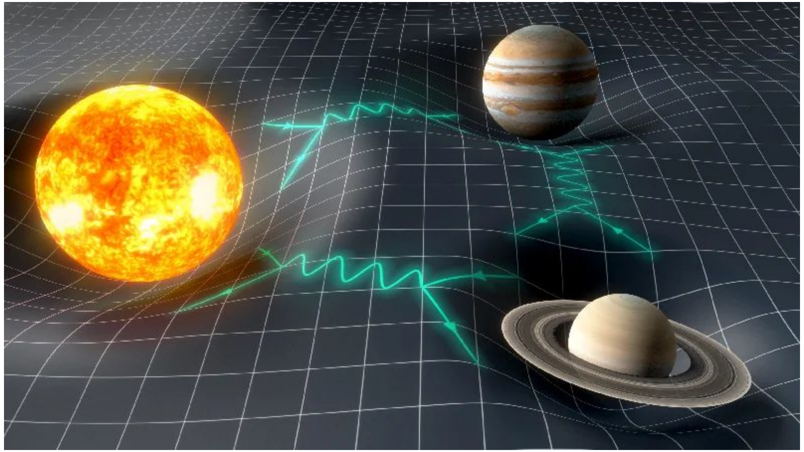
The basic line of questioning

We know that EM waves → photons. Natural to expect that gravitational waves → gravitons.

Question: can we verify this experimentally?



Y. Gominet/Paris Observatory

Part of broader program involving tabletop experiments as well as GW observations. Today, will focus mostly on the GW aspects.

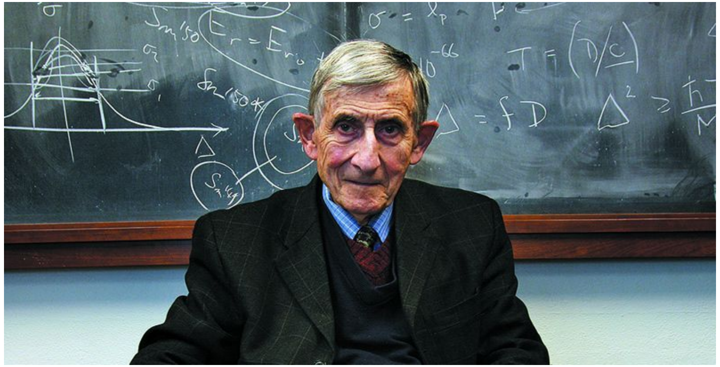
Some reviews:

Technical: Carney, Stamp, Taylor 1807.11494
Popular: Aspelmeyer, Carney, *Physics Today* June 2026

Classic Dyson argument

Can we build a GW detector so sensitive that it will click when a single graviton passes?

Dyson: no.



Is a Graviton Detectable?

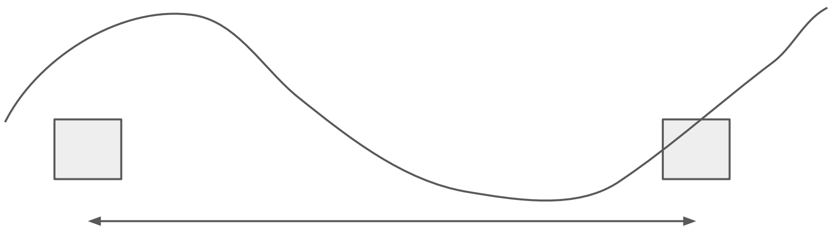
Poincare Prize Lecture

International Congress of Mathematical Physics

Aalborg, Denmark, August 6, 2012

Freeman Dyson, Institute for Advanced Study, Princeton, New Jersey

Classic Dyson argument



Model single graviton as strain field $h \sim L_{pl}\,\omega$

Optimum detection efficiency when $D \sim 1/\omega$

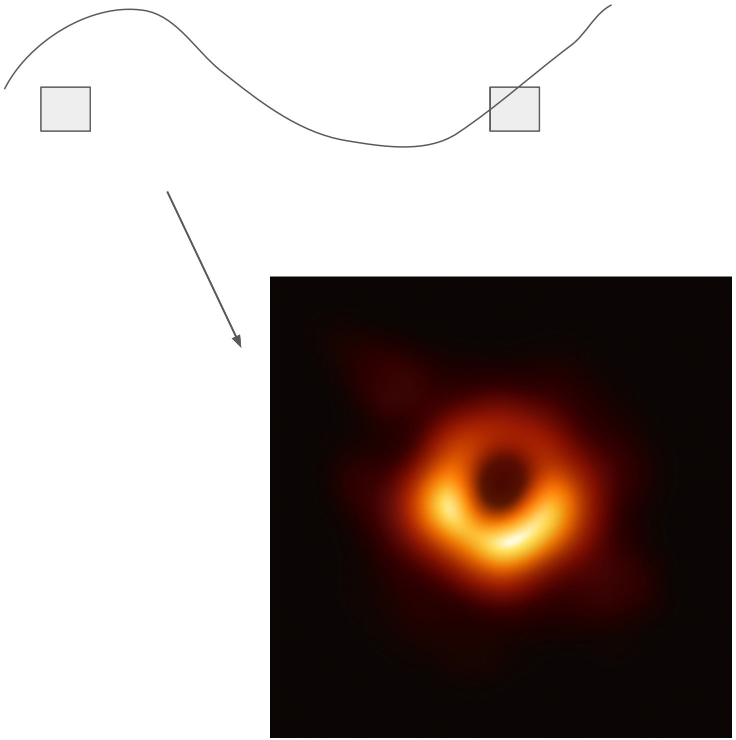
$\rightarrow$ Displacement between mirrors you need to detect is $\Delta x \sim h\,D \sim L_{pl}$

But, Heisenberg uncertainty $\Delta x\,\Delta p \sim M\,\Delta x^2/\Delta t \geq \hbar$

$\rightarrow$ **$D \leq G_N\,M/c^2$**

The detector has to be within its own Schwarzschild radius. If you tried to build a LIGO sensitive to a single graviton strain, it collapses into a black hole!

Classic Dyson argument



I think the core idea of this argument is basically correct, but the real meaning is a little subtle.

To see this, let’s analyze some specific cases and proposals:

1. “Single graviton” detection
2. “Quantum graviton noise” detection
3. More complex ideas: multiple detectors,...

I will argue that all of these are **incapable** of testing the graviton picture of GWs, for the same reason: it’s impossible to build a sufficiently good detector to see meaningful quantum effects due to gravitons.

But at the end I’ll be a little more optimistic…

How to think about this properly

Null hypothesis: gravitons

$$\hat{H}_{\mathrm{Q}} = \hat{H}_{\mathrm{det}} + \hat{H}_h + \int d^3\mathbf{x}\,\hat{h}_{\mu\nu}\hat{T}^{\mu\nu}$$

Quantized metric fluctuations → h = operator

One schematic alternative: “classical” gravity

$$\hat{H}_{\mathrm{SC}} = \hat{H}_{\mathrm{det}} + H_h + \int d^3\mathbf{x}\,h_{\mu\nu}\hat{T}^{\mu\nu}$$

h = classical random variable, drawn from probability distribution P(h)

In detailed realizations, P(h) obeys stochastic/diffusive evolution → effective random Hamiltonian/open system evolution on masses

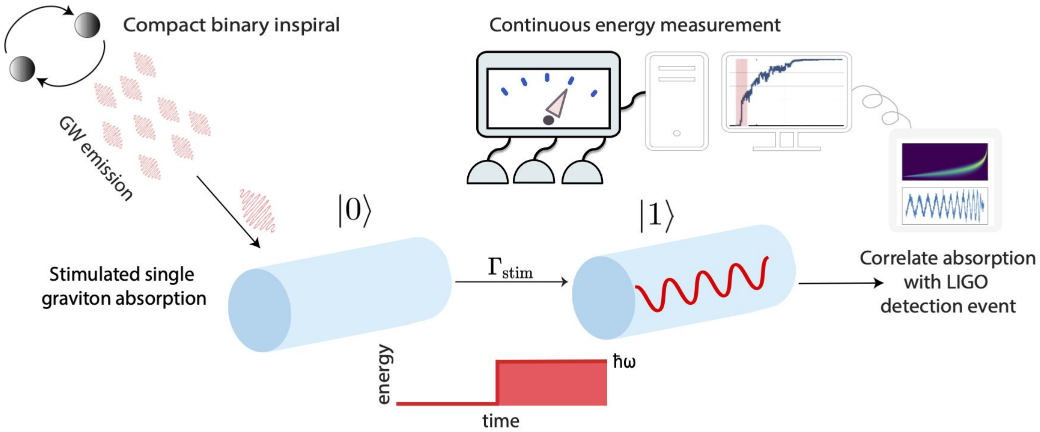
The “correct” question: can we do an experiment that can distinguish between these two options?

Kafri, Taylor, Milburn 2013
Oppenheim, Weller-Davies, et al., 2020 onwards
Tomizuka, Takeda 2026

Possible experiments:

1. **“Single graviton” detection**
2. “Quantum graviton noise” detection
3. More complex ideas: multiple detectors,...

Can we count gravitons?



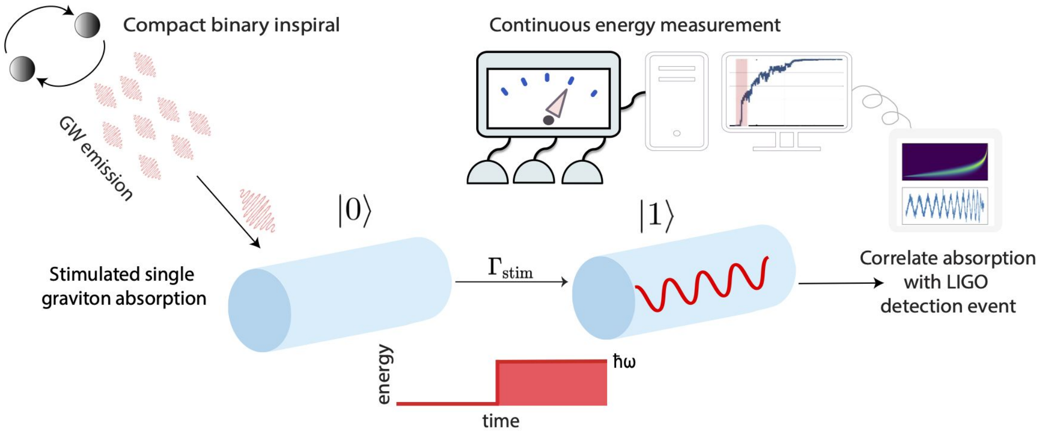
Weber bar:
gravitational wave → vibration → voltage

Fancy quantum Weber bar:
gravitational wave (“gravitons”) → phonon → voltage

This experiment is borderline plausible in the LIGO band

Tobar, Manikandan, Beitel, Pikovski
2308.15440

Can we count gravitons?



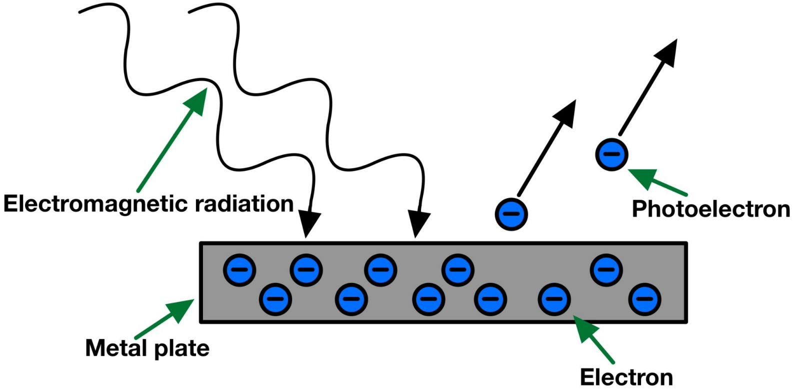
One interpretation: graviton absorbed, converted into phonon

So is this “single graviton detection”? I don’t care. Purely semantic issue.
Instead, the **right** question is: can this distinguish between these:

$$\hat{H}_{\rm Q} = \hat{H}_{\rm det} + \hat{H}_h + \int d^3\mathbf{x}\hat{h}_{\mu\nu}\hat{T}^{\mu\nu} \qquad \Big| \qquad \hat{H}_{\rm SC} = \hat{H}_{\rm det} + H_h + \int d^3\mathbf{x} h_{\mu\nu}\hat{T}^{\mu\nu}$$

Photoelectric version

Frequency = ω



Gap/work function = Δ

Consider either quantum or classical EM fields:

$$\hat{H}_{\mathrm{Q}} = \hat{H}_{\mathrm{det}} + \hat{H}_A + \frac{e}{m}\hat{\mathbf{p}} \cdot \hat{\mathbf{A}}$$

$$\hat{H}_{\mathrm{SC}} = \hat{H}_{\mathrm{det}} + H_A + \frac{e}{m}\hat{\mathbf{p}} \cdot \mathbf{A}$$

Classical, random $P_{cl}(A)$

Textbook perturbation theory leads to identical predictions for excited electron events:

$$\frac{dP}{dt} = \eta \langle I(t) \rangle \Theta(\omega - \Delta)$$

Identical calculation works for perturbative gravity.

→ **"single photon"/"single graviton" detection does not require a photon/graviton explanation!**

Glauber 1963
Carney, Domcke, Rodd arXiv:2308.12988
Carney 2408.00094

Non-classical graviton tests?

So how to distinguish classical/quantum models?

For any quantum state of a field mode, can find a coherent state representation:

$$\rho = \int d\beta P(\beta) |\beta\rangle \langle\beta| , \quad \int d\beta P(\beta) = 1$$

- If $P(\boldsymbol{\beta}) \geq 0$ everywhere → classical ensemble exists, H_{SC} model can reproduce observables
- If $P(\boldsymbol{\beta}) < 0$ somewhere → can find observable that **cannot** be reproduced by H_{SC}

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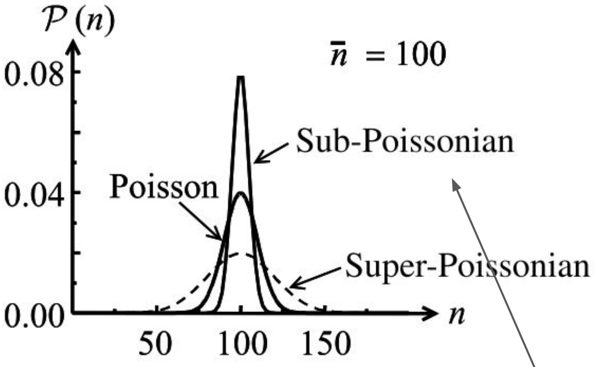


Produced by e.g. squeezed state.

Inconsistent with semiclassical model.

$$\langle \Delta N^2 \rangle - \langle N \rangle = \eta^2 \Delta t^2 \int d\beta P(\beta) \Delta I_\beta^2$$

Observed in photoelectric effect in 1980's

Non-classical graviton tests?

Detecting sub-Poisson statistics requires detector efficiency η = (power absorbed/power incident) ~ O(1)

$$\langle \Delta N^2 \rangle - \langle N \rangle = \eta^2 \Delta t^2 \int d\beta P(\beta) \Delta I_\beta^2$$

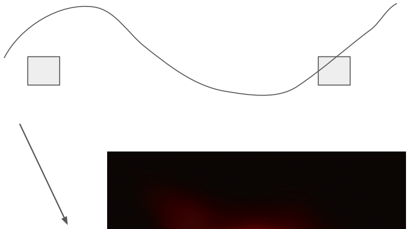
In Weber bar: $\eta \sim 10^{-20}$

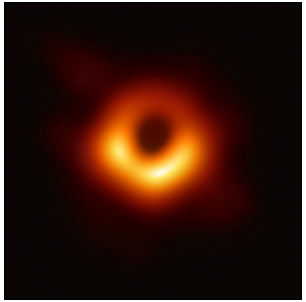
→ Impossibly difficult to detect this. Similar estimate is true for any reasonable detector

Conclusion: any “graviton absorption” signal in a real detector can be explained by a classical GW driving a quantized detector

Carney, Domcke, Rodd arXiv:2308.12988
Carney 2408.00094

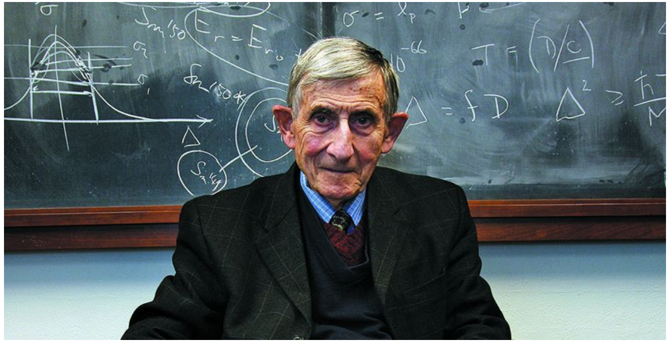
Relation to Dyson’s argument





Dyson: scaling LIGO to be sensitive to a strain of “single graviton” $h \sim L_{pl}\ \omega$ is fundamentally impossible. Why? Quantum noise of detector itself.

Does the single “graviton → phonon” Weber bar detector idea this contradict this? No: it assumes a massive number of incident gravitons, then detector absorbs ~few of these, which overcomes the low detector efficiency $\eta \sim 10^{-20}$



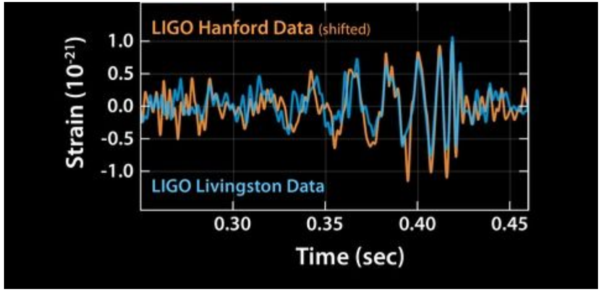
Dyson’s argument is about building something with efficiency $\eta \sim 1$! But that’s what is needed to see a signal that you can *only* explain with gravitons.

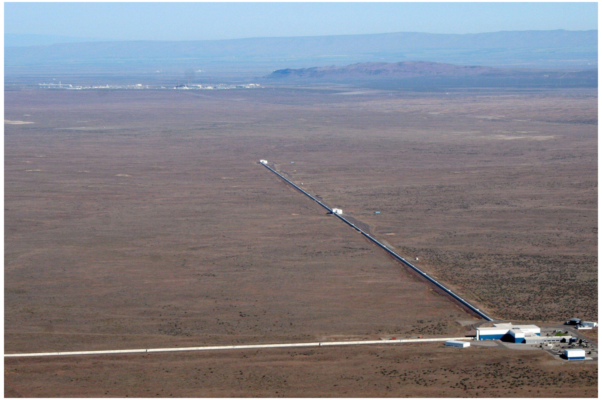
General conclusion: intrinsic quantum noise of any constructible detector >> quantum noise of GW signal

Possible experiments:

1. “Single graviton” detection
2. **“Quantum graviton noise” detection**
3. More complex ideas: multiple detectors,...

Can we detect “quantum graviton noise”?





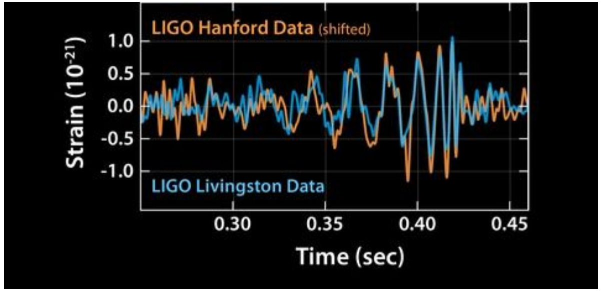
How about some non-classical measurement with a linear detector like LIGO? (as opposed to an absorbing detector)

One idea popularized in Parikh, Wilczek, Zahariade 2005.07211, and repeated in many places: we could look for the noise

Noise ~ <var h(t)>

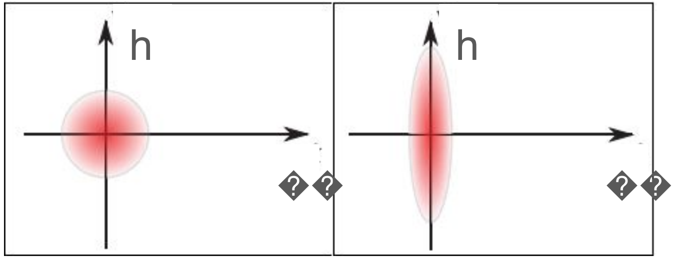
on top of a LIGO-like measurement. Here this noise is supposed to come from the quantum state of GW itself.

Can we detect “quantum graviton noise”?



In particular, many people have suggested looking at squeezed graviton states.

$$S(z) = \exp\left[\frac{1}{2}(zb^\dagger - z^* b)\right], \quad z = re^{i\theta}$$



Gravitational wave in a squeezed state could have highly amplified amplitude fluctuations → would see large “quantum” noise in the detector.

Various sources of these have been proposed (Guerreiro et al 2020 onward, Wilczek et al, Kanno et al 2026, …)

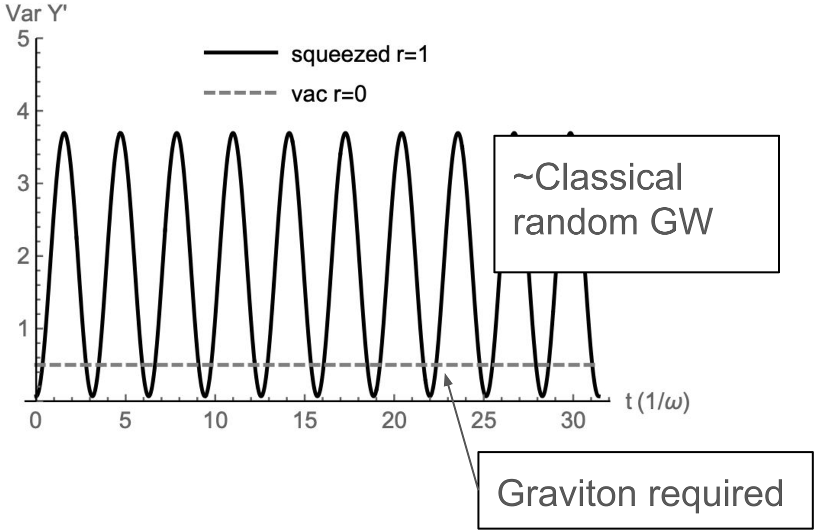
Can we detect "quantum graviton noise"?

However, same core issue arises here.

$$\langle \Delta Y^2 \rangle = \frac{1}{2} + \eta^2 \int d\beta P(\beta)[\mathrm{Re}\ \beta - \langle \mathrm{Re}\ \beta \rangle]^2$$

Optical phase noise measured by interferometer

Vacuum fluctuations of detector itself



Non-classical explanation required if integral term < 0. Produced by squeezed state…

But this term is suppressed by η << 1 → detector's fluctuations dominate any "quantum" signal

Carney, Domcke, Rodd arXiv:2308.12988
Carney 2408.00094

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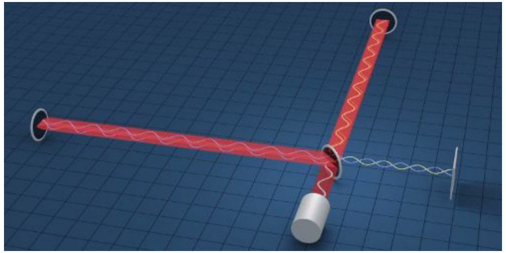
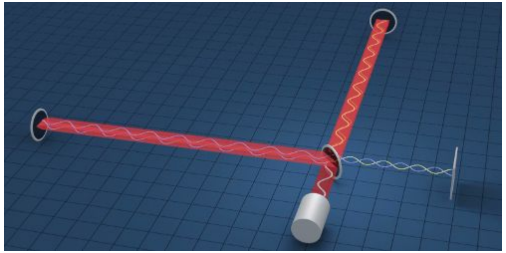
3. **More complex ideas: multiple detectors,...**





Measure some correlated observable? (e.g., noise cross correlations)

Same argument applies: the quantum part of the GW signal will be swamped by the detector’s intrinsic noise.

Let's end on a more positive note…

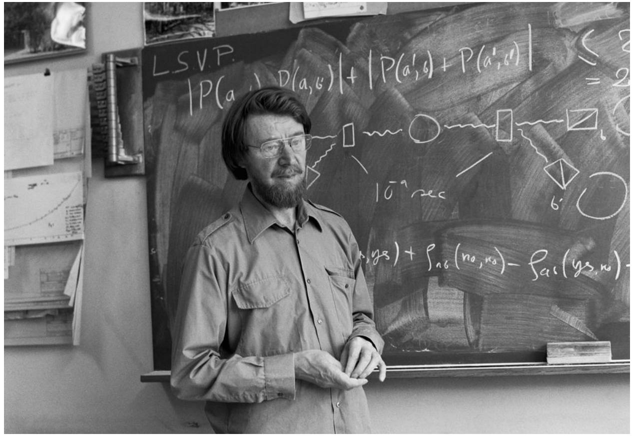
Okay, so GW detectors cannot actually detect a meaningful signature of canonical, perturbative quantization of gravity.

Is all hope lost?

No! Two other good options:

- GW observations can *rule out specific models* where gravity is not quantized as gravitons

- Tabletop experiments can probe the graviton in a somewhat indirect fashion



Excluding anomalous noise

Null hypothesis: gravitons

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Quantized metric fluctuations → h = operator

One schematic alternative: “classical” gravity

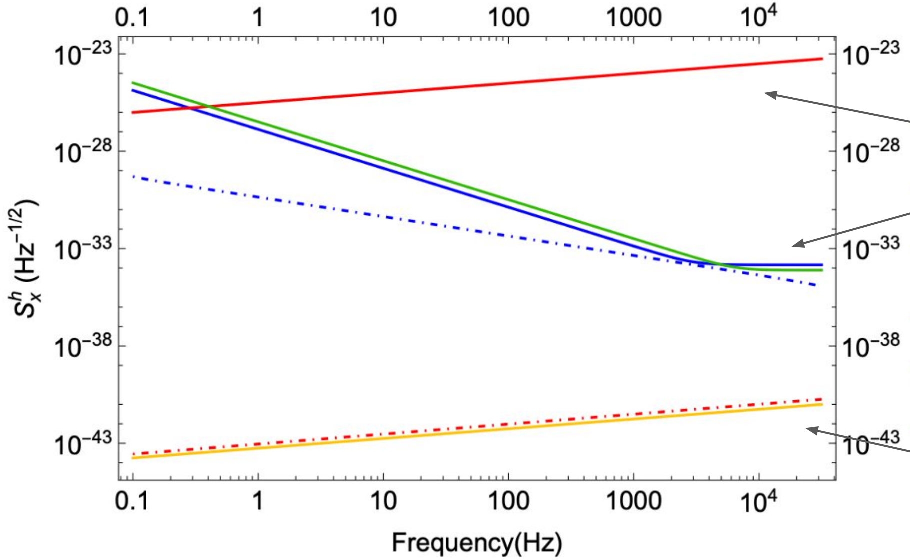
$$\hat{H}_{\mathrm{SC}} = \hat{H}_{\mathrm{det}} + H_h + \int d^3\mathbf{x}\,h_{\mu\nu}\hat{T}^{\mu\nu}$$

h = classical random variable, drawn from probability distribution P(h)

→ **intrinsically noisy gravitational field**

Kafri, Taylor, Milburn 2013
Oppenheim, Weller-Davies, et al., 2020 onwards
Tomizuka, Takeda 2026

Excluding anomalous noise



Some variations of the Oppenheim et al "classical-quantum" (CQ) model, an example of gravity as a classical, stochastic field

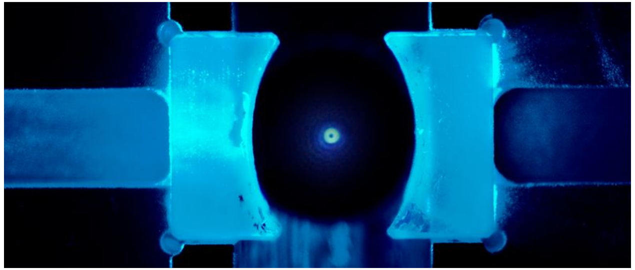
Vacuum fluctuations in perturbative quantum gravity

Hirotani, Matsumura 2603.29230

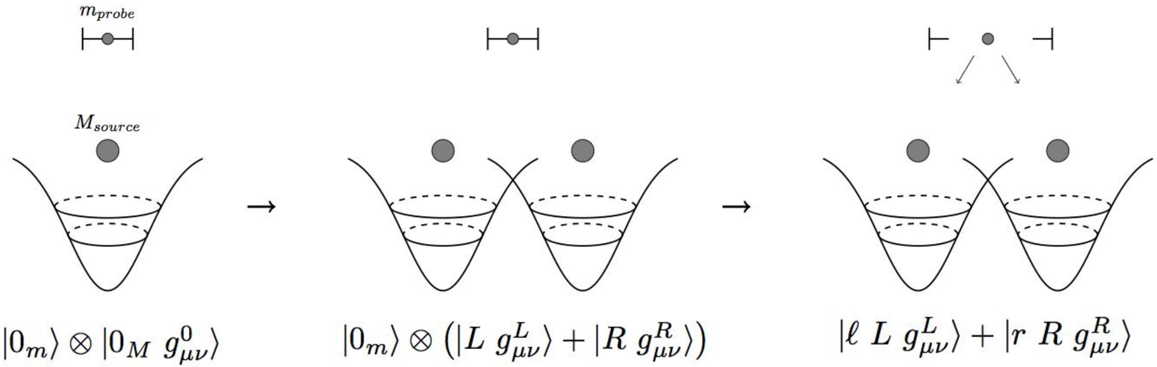
Cf. LIGO's current noise ~ 10^{-23} $Hz^{-1/2}$

Tabletop experiments

Basic goal: prepare a non-classical state of the static gravitational field by preparing a mass. Observe via entanglement generation (or similar observable).



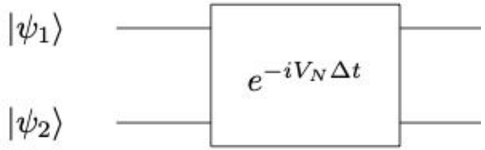
From Markus Aspelmeyer



Kafri, Taylor 2013; Bose et al 2016; + many others
Review: Carney, Stamp, Taylor 1807.11494

Tabletop experiments

Null hypothesis



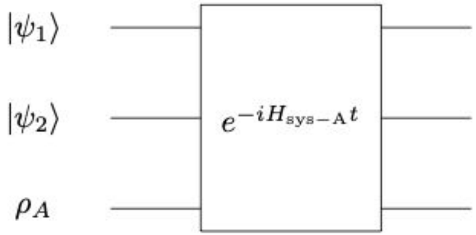
$$\dot{\rho} = i[\hat{H}, \rho]$$

$$\hat{H} = \sum_a \frac{\hat{\mathbf{p}}_a^2}{2m_a} - \frac{1}{2}\sum_{a\neq b} \frac{G_N m_a m_b}{|\hat{\mathbf{x}}_a - \hat{\mathbf{x}}_b|}$$

Null hypothesis: perturbative quantum gravity

Reversible (unitary) interaction, entangling

Alternative hypotheses (can classify these)



$$\dot{\rho} = i[\hat{H}, \rho] + \sum_\alpha \hat{L}_\alpha \rho \hat{L}_\alpha^\dagger - \frac{1}{2}\left\{\hat{L}_\alpha^\dagger \hat{L}_\alpha, \rho\right\}$$

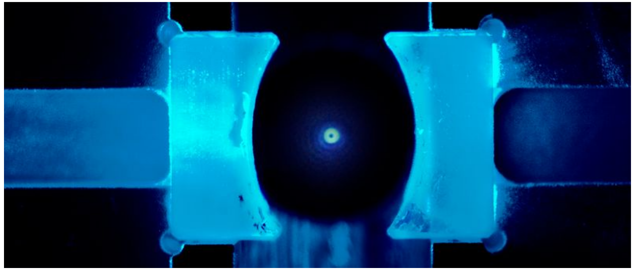
Alternatives:

Not unitary, generally not/less entangling

Full classification: Fabiano, Fujita, Matsumura, Carney 2603.26075

Tabletop experiments

Suppose we observe this gravitational entanglement. This shows that the "h_{00}" metric component can be superposed. What does it imply about the graviton?



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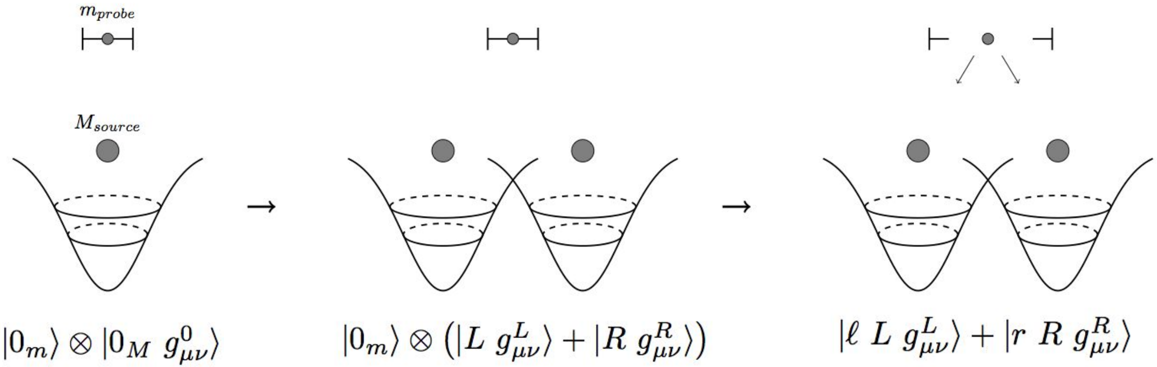


Tabletop experiments

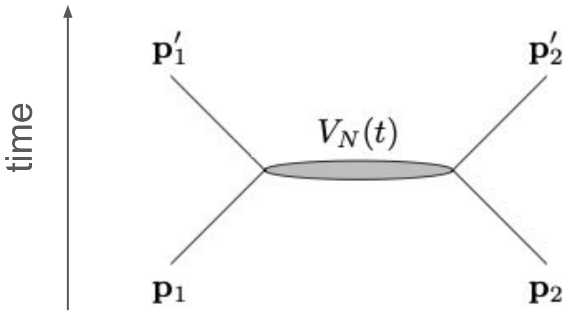
Here is a theorem: any model in which scattering is

- unitary,
- Lorentz invariant, and
- has coherent, entangling $V_N = G\, m_1 m_2/r$ interaction

Necessarily has quantized gravitational radiation.

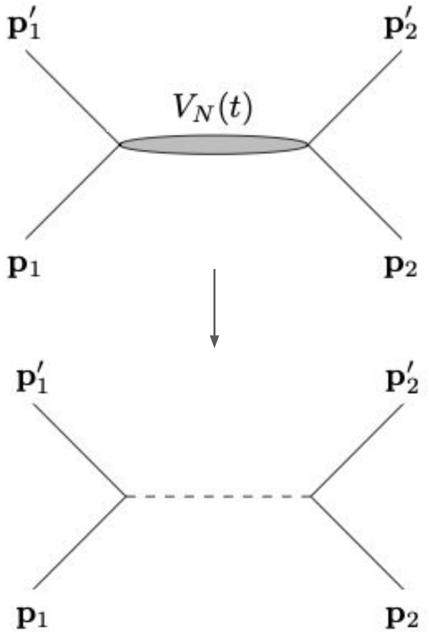
Simple consequence of optical theorem, on next slides.
Basically reverse engineering Weinberg 1960's papers

→ Testing these assumptions = testing if graviton exists



Carney 2108.06320
See also: Belenchia et al 1807.07015; Danielson, Satishchandran, Wald 2112.10798

Lorentz “bootstrap”



Central idea: require non-relativistic scattering amplitudes to be non-relativistic limit of a Lorentz-invariant amplitude

Example: consider 2 → 2 Newtonian scattering

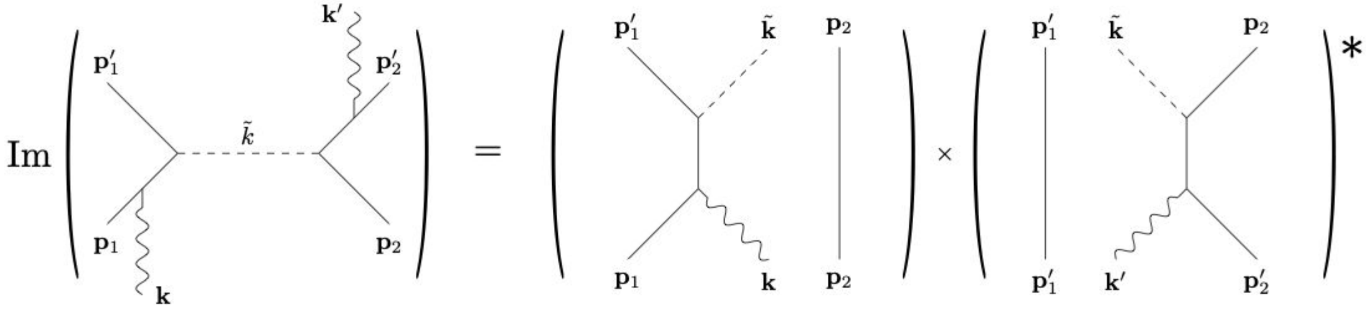
$$\mathcal{M}_{\mathbf{p}_1\mathbf{p}_2\to\mathbf{p}_1'\mathbf{p}_2'} = \langle \mathbf{p}_1'\mathbf{p}_2'|V|\mathbf{p}_1\mathbf{p}_2\rangle = \frac{G_N m^2}{(\mathbf{p}_1' - \mathbf{p}_1)^2 + \mu^2} \longrightarrow \frac{G_N m^2}{(p_1' - p_1)^2 + \mu^2}$$

3-vector

4-vector

Here $\mu \to 0$ is an IR regulator. Unique Lorentz-invariant extension to this order

(Can multiply and add functions which are trivial near the pole $\Delta p^2 = -\mu^2$)



$$1 = S^\dagger S \iff \mathrm{Im}\,\mathcal{M}_{\alpha\to\beta} = \sum_X \mathcal{M}(\alpha \to X)\mathcal{M}^*(\beta \to X)$$

Unitarity + Lorentz invariance requires final-state, quantized gravitational radiation in the sum over X (dashed lines). Argument fixes coupling ~ m $G_N^{1/2}$, but not spin.

→ If you can experimentally determine that the gravitational interaction is unitary and entangling, you prove that the gravitational radiation field is quantized.

Carney 2108.06320

Recap and outlook

1. Can we detect gravitons? Yes, but any signal can also be explained classically
2. Can we detect “quantum noise” from, e.g., some highly squeezed gravitational radiation state? Same statement: yes, but any signal can also be explained classically
3. Can we detect gravitational entanglement generation? Yes, and it would be useful, but the connection to gravitons is even harder (have to verify unitary evolution… how precisely?)

Big open questions in this field:

1. Can we construct a truly self-consistent and viable “non graviton” model that reproduces full GR in appropriate limits?
2. Can we classify all such models?
3. Can we think of some smarter experiments???